FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

MIDTERM

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10-05-2026	10-09-2026
Data Performed	Date Submitted

List the steps that you perform to come up with the program.

- Customize the output design, or the layout of my slambook.
- Ask for the user input (name, email, etc…).
- Store them and outputs them using a loop.

Paste the source code and explain the functionalities and take a screenshot of the final output.

```
global _start

section .data
   cls        db 0x1b, "[2J", 0x1b, "[H", 0x1b, "[44;37m", 0
   cls_len    equ $ - cls

   top_frame   db
"╔═══════════════════════════════════════════════════════════════════════
═══════╗", 10, 0
   top_frame_l equ $ - top_frame

   banner     db "║        * MY INTERACTIVE DIGITAL SLAMBOOK *         ║", 10
           db
"╠═══════════════════════════════════════════════════════════════════════
═══════╣", 10, 0
   banner_len  equ $ - banner

   side_l     db "║   ", 0
   side_l_len  equ $ - side_l

   raw_nl     db 10, 0
   raw_nl_len  equ $ - raw_nl

   div_frame   db
"╠═══════════════════════════════════════════════════════════════════════
═══════╣", 10, 0
   div_frame_l equ $ - div_frame

   bot_frame   db
"╚═══════════════════════════════════════════════════════════════════════
═══════╝", 10, 0
   bot_frame_l equ $ - bot_frame

   s_about    db "║              ✦ [ ABOUT ME PROFILE ] ✦              ║", 10, 0
```

```
    s_about_l   equ $ - s_about

    s_firsts    db "╠═══════════ ✦ [ LIFE MY FIRSTS ] ✦ ═══════════╣", 10, 0
    s_firsts_l  equ $ - s_firsts

    s_faves     db "╠═══════════ ✦ [ PERSONAL FAVES ] ✦ ═══════════╣", 10, 0
    s_faves_l   equ $ - s_faves

    s_motto     db "╠═══════════ ✦ [ HOBBIES & LIFE MOTTO ] ✦ ═══════════╣", 10, 0
    s_motto_l   equ $ - s_motto

    lbl_name    db "Name:         ", 0
    lbl_name_l  equ $ - lbl_name
    lbl_email   db "Email:        ", 0
    lbl_email_l equ $ - lbl_email
    lbl_blog    db "Blog/Website: ", 0
    lbl_blog_l  equ $ - lbl_blog

    lbl_achieve db "First Achievement: ", 0
    lbl_achieve_l equ $ - lbl_achieve
    lbl_risk    db "First Risk Taken:  ", 0
    lbl_risk_l  equ $ - lbl_risk

    lbl_colors  db "Favorite Colors:   ", 0
    lbl_colors_l equ $ - lbl_colors
    lbl_food    db "Favorite Food:     ", 0
    lbl_food_l  equ $ - lbl_food

    lbl_hobby   db "Hobby Activity:    ", 0
    lbl_hobby_l equ $ - lbl_hobby
    lbl_motto   db "Life Motto:        ", 0
    lbl_motto_l equ $ - lbl_motto

    summary_h   db "╠═══════════ ✦ VERIFYING STORED DATA PROFILE ✦ ═══════════╣", 10
            db
"╠═══════════════════════════════════════════════════════════════════════════════╣", 10, 0
    summary_hl  equ $ - summary_h

    done_msg    db 10, "Press [ENTER] to save record and close app...", 10, 0
    done_msg_l  equ $ - done_msg
```

```
section .bss
   buf_name    resb 32
   buf_email   resb 32
   buf_blog    resb 32
   buf_achieve resb 32
   buf_risk    resb 32
   buf_colors  resb 32
   buf_food    resb 32
   buf_hobby   resb 32
   buf_motto   resb 32

section .text
_start:
   mov eax, 4
   mov ebx, 1
   mov ecx, cls
   mov edx, cls_len
   int 0x80

   mov eax, 4
   mov ecx, top_frame
   mov edx, top_frame_l
   int 0x80

   mov eax, 4
   mov ecx, banner
   mov edx, banner_len
   int 0x80

   mov eax, 4
   mov ecx, s_about
   mov edx, s_about_l
   int 0x80

   call print_side_borders
   mov eax, 4
   mov ecx, lbl_name
   mov edx, lbl_name_l
   int 0x80
   mov ecx, buf_name
   call get_input
```

```
call print_side_borders
mov eax, 4
mov ecx, lbl_email
mov edx, lbl_email_l
int 0x80
mov ecx, buf_email
call get_input

call print_side_borders
mov eax, 4
mov ecx, lbl_blog
mov edx, lbl_blog_l
int 0x80
mov ecx, buf_blog
call get_input

mov eax, 4
mov ecx, div_frame
mov edx, div_frame_l
int 0x80

mov eax, 4
mov ecx, s_firsts
mov edx, s_firsts_l
int 0x80

call print_side_borders
mov eax, 4
mov ecx, lbl_achieve
mov edx, lbl_achieve_l
int 0x80
mov ecx, buf_achieve
call get_input

call print_side_borders
mov eax, 4
mov ecx, lbl_risk
mov edx, lbl_risk_l
int 0x80
mov ecx, buf_risk
call get_input
```

```
mov eax, 4
mov ecx, div_frame
mov edx, div_frame_l
int 0x80

mov eax, 4
mov ecx, s_faves
mov edx, s_faves_l
int 0x80

call print_side_borders
mov eax, 4
mov ecx, lbl_colors
mov edx, lbl_colors_l
int 0x80
mov ecx, buf_colors
call get_input

call print_side_borders
mov eax, 4
mov ecx, lbl_food
mov edx, lbl_food_l
int 0x80
mov ecx, buf_food
call get_input

mov eax, 4
mov ecx, div_frame
mov edx, div_frame_l
int 0x80

mov eax, 4
mov ecx, s_motto
mov edx, s_motto_l
int 0x80

call print_side_borders
mov eax, 4
mov ecx, lbl_hobby
mov edx, lbl_hobby_l
int 0x80
mov ecx, buf_hobby
```

```
call get_input

call print_side_borders
mov eax, 4
mov ecx, lbl_motto
mov edx, lbl_motto_l
int 0x80
mov ecx, buf_motto
call get_input

mov eax, 4
mov ecx, bot_frame
mov edx, bot_frame_l
int 0x80

mov ecx, buf_name
call clean_input
mov ecx, buf_email
call clean_input
mov ecx, buf_blog
call clean_input
mov ecx, buf_achieve
call clean_input
mov ecx, buf_risk
call clean_input
mov ecx, buf_colors
call clean_input
mov ecx, buf_food
call clean_input
mov ecx, buf_hobby
call clean_input
mov ecx, buf_motto
call clean_input

mov eax, 4
mov ebx, 1
mov ecx, cls
mov edx, cls_len
int 0x80

mov eax, 4
mov ecx, top_frame
```

```
mov edx, top_frame_l
int 0x80

mov eax, 4
mov ecx, summary_h
mov edx, summary_hl
int 0x80

call print_side_borders
mov eax, 4
mov ecx, lbl_name
mov edx, lbl_name_l
int 0x80
mov eax, 4
mov ecx, buf_name
mov edx, 32
int 0x80
call print_newline

call print_side_borders
mov eax, 4
mov ecx, lbl_email
mov edx, lbl_email_l
int 0x80
mov eax, 4
mov ecx, buf_email
mov edx, 32
int 0x80
call print_newline

call print_side_borders
mov eax, 4
mov ecx, lbl_colors
mov edx, lbl_colors_l
int 0x80
mov eax, 4
mov ecx, buf_colors
mov edx, 32
int 0x80
call print_newline

call print_side_borders
```

```
    mov eax, 4
    mov ecx, lbl_motto
    mov edx, lbl_motto_l
    int 0x80
    mov eax, 4
    mov ecx, buf_motto
    mov edx, 32
    int 0x80
    call print_newline

    mov eax, 4
    mov ecx, bot_frame
    mov edx, bot_frame_l
    int 0x80

    mov eax, 4
    mov ecx, done_msg
    mov edx, done_msg_l
    int 0x80

    mov ecx, buf_name
    call get_input

    mov eax, 1
    xor ebx, ebx
    int 0x80

print_side_borders:
    mov eax, 4
    mov ebx, 1
    mov ecx, side_l
    mov edx, side_l_len
    int 0x80
    ret

print_newline:
    mov eax, 4
    mov ebx, 1
    mov ecx, raw_nl
    mov edx, raw_nl_len
    int 0x80
    ret
```

```
get_input:
    mov eax, 3
    mov ebx, 0
    mov edx, 32
    int 0x80
    ret

clean_input:
    mov esi, ecx
.loop:
    mov al, [esi]
    cmp al, 10
    je .found
    cmp al, 0
    je .done
    inc esi
    jmp .loop
.found:
    mov byte [esi], 0
.done:
    ret
```